Prompt Engineering for Generative AI

COMP4146/7125 Prompt Engineering for Generative AI,

Semester 2, 2025/26

Dr. Shichao Ma

The popularity of Prompt Engineering

- LinkedIn reports that **3× more C-suite leaders** are adding AI skills (including **prompt engineering**) to their profiles vs. two years prior, and frames **AI literacy** as a top leadership skill priority.
  - Published on Apr 29, 2025
  - https://news.linkedin.com/2025/ai-adoption-starts-at-the-top--3x-more-c-suites-on-linkedin-are-
- ***Prompt engineering is entering the executive mainstream (not just engineers).***

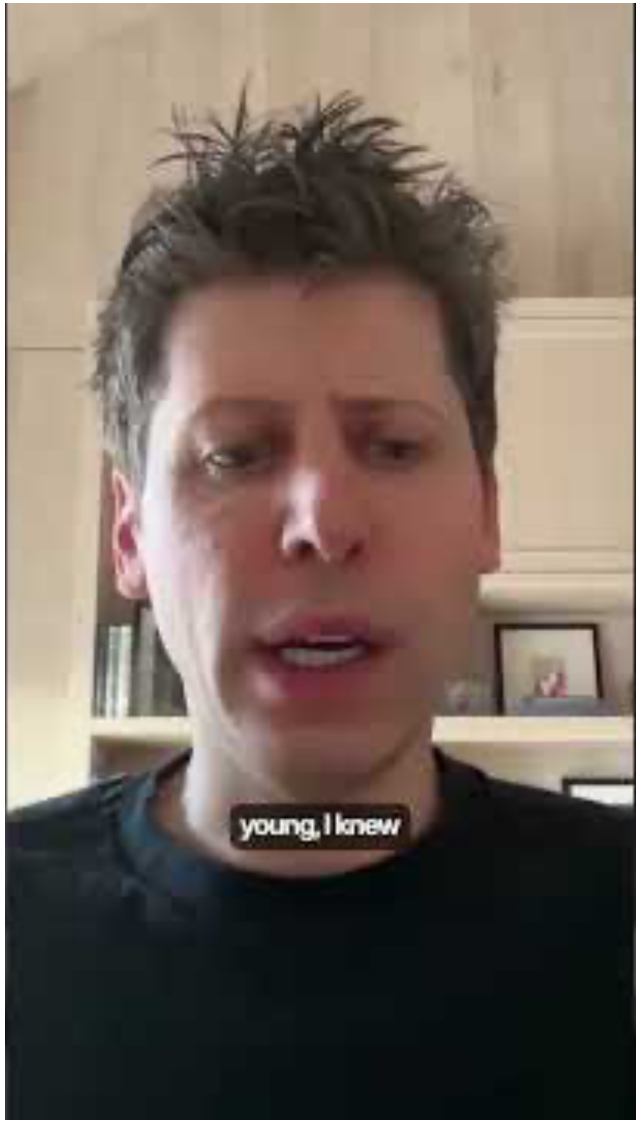
The popularity of Prompt Engineering

- Microsoft + LinkedIn Work Trend Index reports explosive growth in members adding AI skills to profiles and shows that **AI mentions in job posts increase applications** (a proxy for popularity + market pull).
  - Released on May 8, 2024
  - https://news.microsoft.com/source/2024/05/08/microsoft-and-linkedin-release-the-2024-work-trend-index-on-the-state-of-ai-at-work/
- ***AI/prompting is 'resume-worthy' and affects job-market behavior.***

The popularity of Prompt Engineering

- Washington Post made a coverage of employers increasingly seeking "**AI literate**" candidates across roles, reflecting broad normalization of AI tool use (where prompting is a core practical capability)
  - Released on Aug 30, 2025
  - https://www.washingtonpost.com/business/2025/08/30/job-skills-ai-literacy/
- ***Prompting is becoming a baseline workplace literacy.***

The popularity of Prompt Engineering



Sam Altman on: Is Prompt Engineering a REAL Job?

So … what is "Prompt Engineering" exactly? 🤔

Video source:
https://www.youtube.com/shorts/onh5FlzGB8I?feature=share

“Prompt”, as a word

prompt
/prɒm(p)t/

verb

1. (of an event or fact) cause or bring about (an action or feeling).
   "the violence prompted a wave of refugees to flee the country"

   Similar: give rise to | bring about | cause | occasion | result in

2. encourage (a hesitating speaker) to say something.
   "‘And the picture?’ he prompted"

   Similar: remind | cue | give someone a cue | help out | coach | feed

noun

1. an act of encouraging a hesitating speaker.
   "with barely a prompt, Barbara talked on"

2. COMPUTING
   a word or symbol on a screen to show that the system is waiting for input.

adjective

done without delay; immediate.
"she would have died but for the prompt action of two ambulancemen"

Similar: quick | swift | rapid | speedy | fast | direct | immediate

adverb BRITISH ENGLISH

exactly (with reference to a specified time).
"I set off at three-thirty prompt"

Similar: exactly | precisely | sharp | on the dot | dead | dead on

“Prompt”, as a word

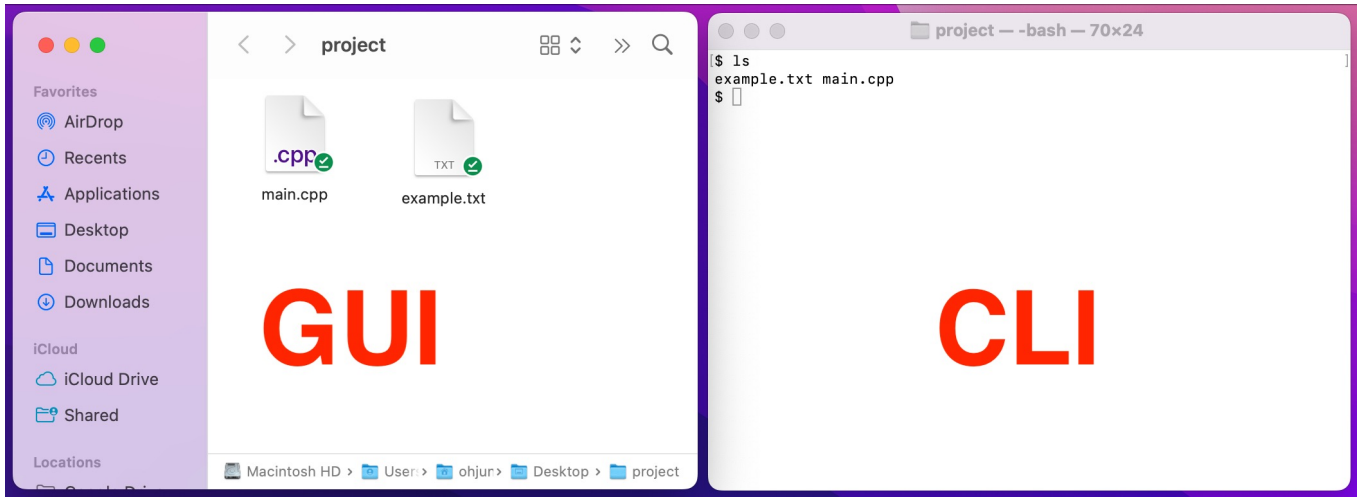
- An act of encouraging a hesitating speaker.
  - Example: “with barely a prompt, Babara talked on”

Prompt to a computer

- Now imagine you are talking to a computer, instead of Barbara, in natural languages.
- That is “prompting” in the scope of Computer Science.

Why it is important?

- There are multiple ways to interact with a computer.
- **CLI (Basic instructions) => GUI (What you see is what you get) => Natural Language (Language is what make us human)**
  - Closer to human habits.
  - More powerful and flexible.
- For each transition, there will be many opportunities



How about “Engineering”?

- **Engineering** is the practice of applying scientific, mathematical, and technical principles to design, build, and maintain structures, machines, systems, and processes.
- It is **a problem-solving discipline** that uses knowledge from natural sciences **to create solutions** for real-world needs, from the infrastructure of cities to the technology in our pockets.
- Key branches include chemical, civil, electrical, and mechanical engineering, each focusing on specific areas like materials, infrastructure, electronics, and machinery.

What engineering involves

- **Problem-solving:**
  - Identifying challenges and creating solutions.
- **Design:**
  - Planning and creating blueprints for new products or systems.
- **Building:**
  - Constructing the designed solutions.
- **Testing and analysis:**
  - Ensuring that the final product functions correctly and safely.
- **Improving existing systems:**
  - Enhancing existing structures or processes for better efficiency and functionality.

Prompt engineering and generative models

- Some history
  - Early NLP systems were rule-based (hand-written grammars, templates).
  - Modern systems use large neural generative models trained on massive text corpora.
  - With generative models, the input text itself becomes a powerful control mechanism → **prompting**.
- **Prompt engineering only makes sense in the context of generative models**
  - Generative models produce new content (text, code, images) by predicting likely continuations.
  - Therefore, they can be steered by instructions, examples, constraints, tone, and formatting in the prompt.

It differs from non-generative systems

- Many programs are **deterministic tools**:
  - You can't "prompt" the **Calculator app** to "be more creative" with results.
  - **A binary classifier** (e.g., dog vs. cat) outputs a label/probability, not an open-ended response.
- These systems can't be "encouraged to say something" because they don't generate language; they **execute** a fixed function.
- Prompt engineering is essentially **interface design for probabilistic text generation**:
  - **specify task + context + constraints + desired output format**
  - to reliably guide a generative model's behavior.

Beyond the generative model

- **Generative models = capabilities ("the brain")**
  - Generate text/code/images by learning patterns in data (next-token / next-step prediction)
  - Strong at reasoning, drafting, summarizing, planning
  - Typically, stateless and non-action-taking unless connected to tools
- **AI agents = systems ("brain + hands + goals")**
  - Goal-directed loop: plan → act → observe → revise
  - Uses tools/APIs (search, databases, email, code execution, enterprise systems)
  - Adds state & memory (context, history, preferences, task progress)
  - Includes controls (validation, permissions, logging, human approval when needed)

Beyond the generative model

- **Prompt engineering** may help (but of course, itself isn't sufficient)
  - Prompts specify role, goals, constraints, tool-use rules, output formats
- The reliability of AI agent usually comes from orchestration + retrieval + validation + monitoring
  - Prompting → Tool Use → Feedback Loop → Reliability
- An AI agent needs other levers: workflow design, fine-tuning, policies/guardrails

Meta acquires Manus, a Singaporean based AI startup for over US$2 billion

The acquisition is one of the largest US-Asia tech deals in history. Manus is a Singapore-based company with Chinese founders. The company will continue operating from Singapore, but exit all Chinese ownership.

The company will integrate its AI agents into Meta products.

About this course

- It's not about finding the "magic word", the goal of this course is to equip you with the theory, techniques, tips, and tricks you need to master prompt engineering and build successful LLM applications.
  - Explain LLM basics, how they work, and how to build apps with them.
  - Teach essential prompt-engineering: getting context, prioritizing, packing, and templating.
  - Cover advanced methods: building LLM workflows and evaluating model performance.
- Delivered by **Lectures** + **Labs**, assessed by **Group project** (more details on the later pages)

About this course

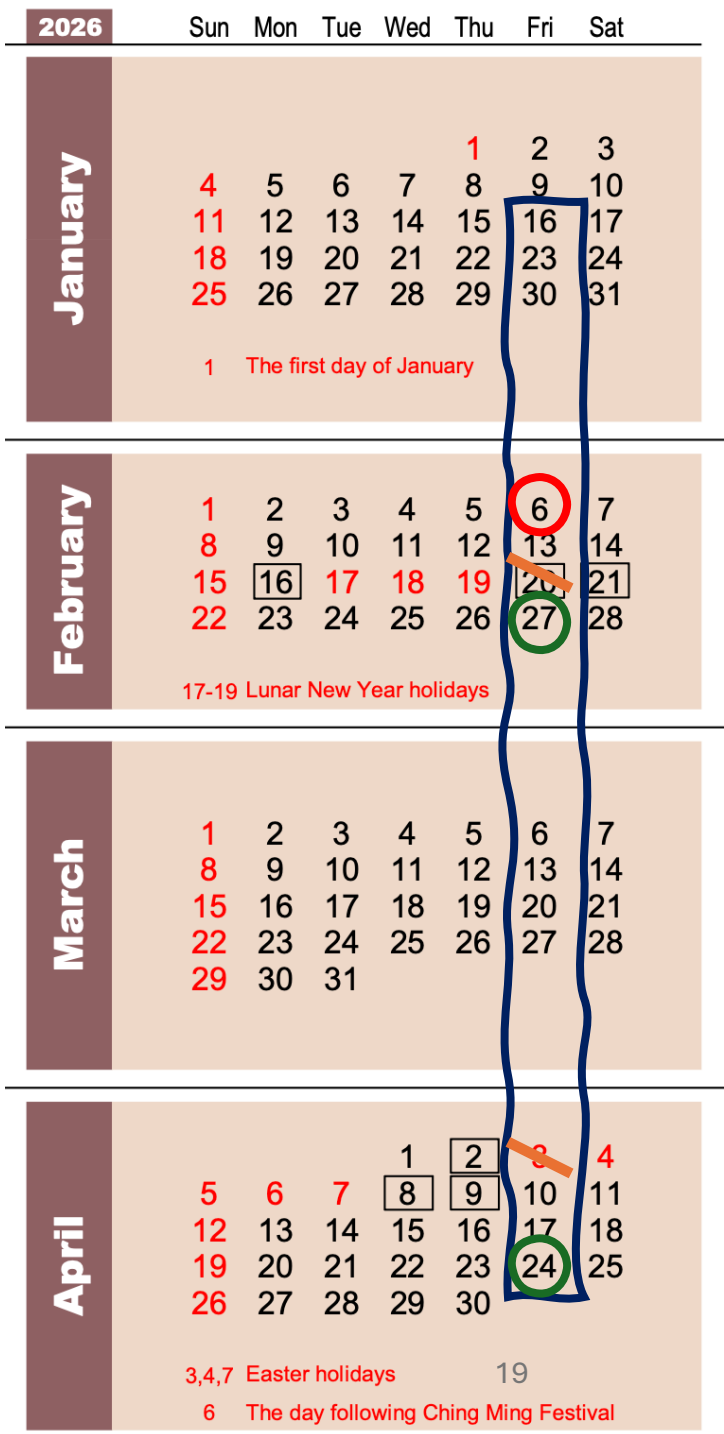
- **Some prerequisite modules:**
  - COMP7015 Artificial Intelligence
  - COMP7025 Artificial Intelligence for Digital Transformation
  - COMP3117 Applied Artificial Intelligence
  - Or other AI-related courses
- **You are expected to know things like:**
  - What is a generative model
  - What is a Transformer
  - What is Softmax
  - What is Transfer Learning, etc.

About this course

- Prompt Engineering is an emerging, fast-evolving field, and this is a new course
  - You are pioneers 🚀
  - We may adjust topics, pacing, or the structure of the course based on progress and feedback.
- A small amount of material in the first few lectures may overlap with COMP7015

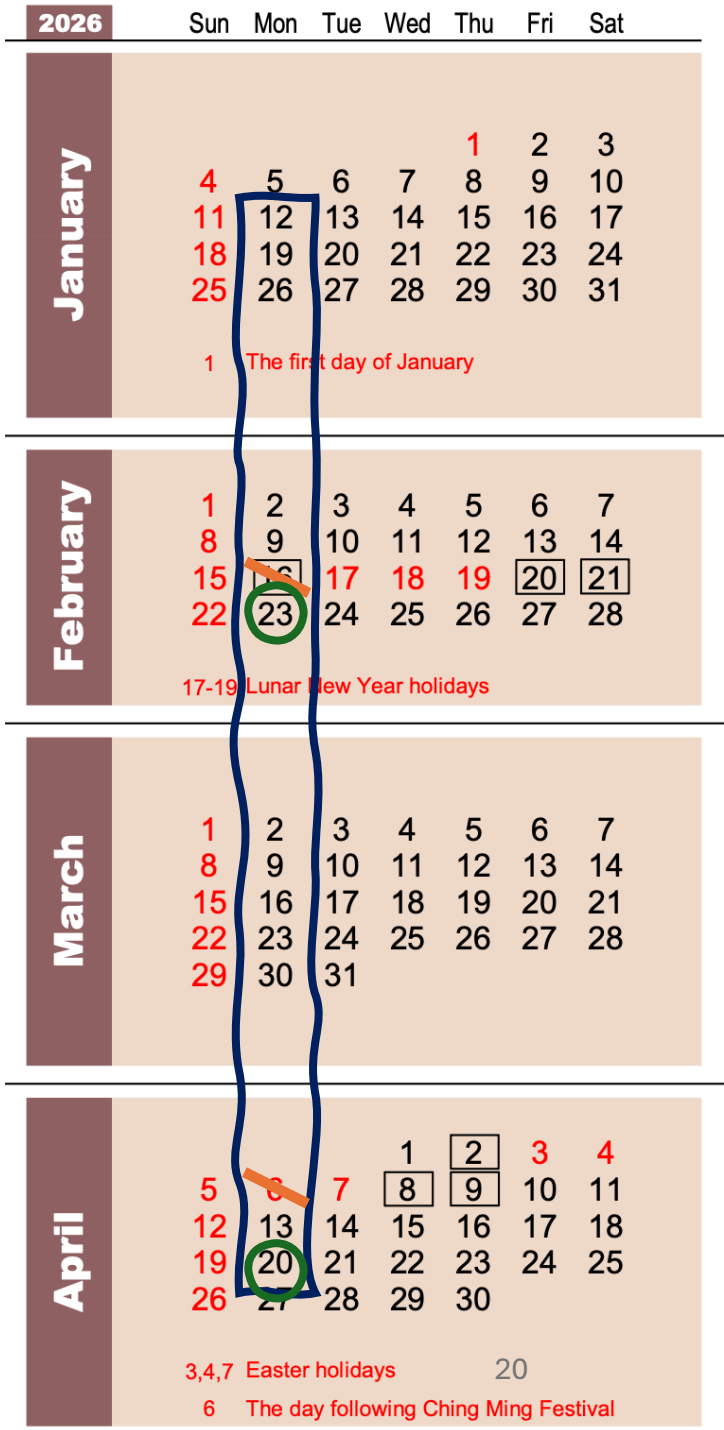
Course Schedule: Lectures

- Every **Friday, from 11:30 – 13:20, @WLB104**.
  - **Except 20th Feb (CNY Holiday) and 3rd Apr (Easter Holiday)**
  - **27th Feb:** In-class quiz @WLB104
  - **24th Apr:** Group project presentation
  - **The lecture scheduled for 6th February will be delivered via Zoom**, because WLB104 will be used for the university's 70th anniversary events. The details will be announced on Moodle due course.



Course Schedule: Lab Sessions

- Every **Monday, from 13:30 – 14:20, @FSC801CD & FSC901CD.**
  - **Except 16th Feb (CNY Holiday) and 6th Apr (Easter Holiday)**
  - **23rd Feb & 20th Apr:** Q&A sessions (Optional)
  - Apart from the introduction session today, **there will be 10 lab sessions in total, covering from Topic 1 to Topic 10.**



Course Content

- **11 Topics in total**
- **Part I:** Prompt Engineering and GenAI Fundamentals
  - About “What is it”
  - **Topic 1 ~ 3**
- **Part II:** Core Prompt Engineering Techniques
  - About “How it works” and “How to utilize it”
  - **Topic 4 ~ 9**
- **Part III:** Beyond the Prompt Engineering
  - **Topic 10 ~ 11**

Assessment

- **20%:**
  - Lab sheets * **10**, **2 marks** for each
  - Submit after the corresponding lab sessions
  - It is an assessment on your engagement of the course
- **30%:**
  - **10% - In-class quiz (27th Feb)**
    - Similar format with the quizzes in 7015 and 7180
  - **20% - Group project**
    - Details will be announced after the quiz
- **50%:** Final examination

Instructor

- **Dr. Shichao Ma**
  - BSc, MSc, PhD
- Research Interest:
  - Deep learning
  - Digital Health
  - Artificial Intelligence in Medicine
  - Biomedical Data Science
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Prompt: *Transform the provided photo into a photorealistic corporate headshot in the distinct style of Apple leadership portraits. The subject should have a confident and approachable expression with a subtle, knowing smile. The lighting should be soft and professional, mimicking studio quality with a gentle key light from the side and soft fill light to minimize harsh shadows. The background should be a clean, minimalist, and slightly out-of-focus light gray or off-white wall. The composition should be a classic head-and-shoulders shot with the subject looking directly into the camera. The attire should be a simple, high-quality dark-colored crewneck sweater. The overall aesthetic must be clean, modern, sharp, and high-resolution.*

AI generated with the prompt on the right.

Teaching Assistants

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